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9

Identity Formation in Emerging Adulthood:

Looking Back

The present chapter summarizes the main findings and tries to formulate an identity formation model that could inspire future researchers who are interested in identity development in emerging adulthood. This model is related to the vast literature on identity formation and related topics. Findings will be summarized according to the 5 major research aims of this dissertation identified in Chapter 1: (a) development of a theoretical framework for identity formation in emerging adulthood, (b) external validation and correlates of this new identity framework, (c) a view on the antecedents and consequences of this identity formation model from a longitudinal perspective, (d) general longitudinal trends, and (e) identification of distinct identity development trajectory classes.

Aim 1: Development of a Theoretical Model for Identity Research

Although identity formation is generally believed to start in adolescence, it certainly intensifies in emerging adulthood. In Chapter 1, we defined emerging adulthood as the age of identity explorations, instability, continual revisions, and possibilities (Arnett, 2004). Hence, a comprehensive framework is needed to capture the identity dynamics in this period of life. Chapters 3 to 5 aimed at developing an integrative model to capture these identity dynamics in emerging adulthood. To that aim, we drew on two leading traditions in the identity field to date, as explained in Chapter 1, and tried to integrate them by focusing on four identity dimensions. In short, the first tradition focused on the formation of identity relevant commitments (e.g., Marcia, 1966, 1980, 1993; Waterman, 1982, 1999a) and distinguished between the exploration of various identity alternatives (or exploration in breadth) and the actual making of identity choices (or commitment making). The second tradition focused on the evaluation of the identity choices one has made (e.g., Bosma, 1985; Bosma & Kunnen, 2001; Grotevant, 1987; Kerpelman et al., 1997; Meeus, 1996) and distinguished between exploring and thinking about current commitments (or exploration in depth) and the degree to which one identifies with and feels certain about these choices (or identification with commitment).

Associations Among the Four Dimensions and Link With the Status Literature

To date, we measured the four identity dimensions of our model (using the Ego Identity Process Questionnaire (EIPQ; Balistreri et al., 1995) and the Utrecht-Groningen Identity Development Scale (U-GIDS; Meeus, 1996)) in four independent samples of college students, one of them being the sample of L-TIDES. Three samples consisted of psychology and educational sciences students (N = 565, N = 199, and N = 399, respectively) with a predominance of females, and one sample consisted of agri- and horticulture students (N = 210)

with a predominance of males. In all four samples, the concurrent correlations were in line with one another and, hence, the structural relations described in our identity formation model were validated across samples. Commitment making and exploration in breadth were negatively correlated in all four samples (*r*s were -.44, -.31, -.31, and -.15; all *p*s < .05). Identification with commitment and exploration in depth were positively interrelated (*r*s were .54, .43, .41, and .50; all *p*s < .001). Further, commitment making and identification with commitment were positively interrelated (*r*s were .42, .32, .33, and .24; all *p*s < .001). Finally, both exploration dimensions were positively interrelated (*r*s were .23, .17, .15, and .04) with the last correlation failing to reach significance ($p > .05$). Chapter 4 used these dimensions in a typological approach to identity. Not only were Marcia's identity statuses more clearly defined using four rather than two identity dimensions, but a new identity status (i.e., carefree diffusion) also emerged from the data. This new identity status may be typical for the emerging adult period – characterized as a period granting a large amount of freedom to adolescents in their search for an inner identity - because this status is characterized by an avoidant orientation with respect to identity issues, coupled with a carefree attitude as expressed in relatively high scores on well-being (van Hoof, 1999b).

Developing the Internal Logic of the Identity Formation Model

Our identity formation model was put to a longitudinal test in Chapter 5 and is presented graphically in Figure 9.1. Simply stated, the model posits that exploration in breadth might give rise to commitment making (i.e., the commitment formation cycle; Path A in Figure 9.1) which, in turn, could activate exploration in depth of those current commitments (Path B in Figure 9.1). Finally, partially as an outcome of this process, the degree of identification with these commitments increases or decreases (i.e., the commitment evaluation cycle; Path C in Figure 9.1). An (ideal case) example demonstrates the internal logic of the model. Consider a late adolescent who starts at college. After he or she explored various possibilities for topics to major in through reading flyers or talking with others (exploration in breadth), he or she will choose one specific major (commitment making). The fact that he or she chose a major does not automatically imply that the identity formation process ceases. He or she will probably continue to gather information (exploration in depth) in order to evaluate the choice being made. Gathering information about that specific choice can lead to a growing conviction that the chosen major is the right one (hence, identification with that major will strengthen) or, conversely, that the chosen major is not the right one (hence, identification with that major will

weaken). Thus, the dynamic interplay between both identity dimensions constitutes the evaluative aspect of identity formation.

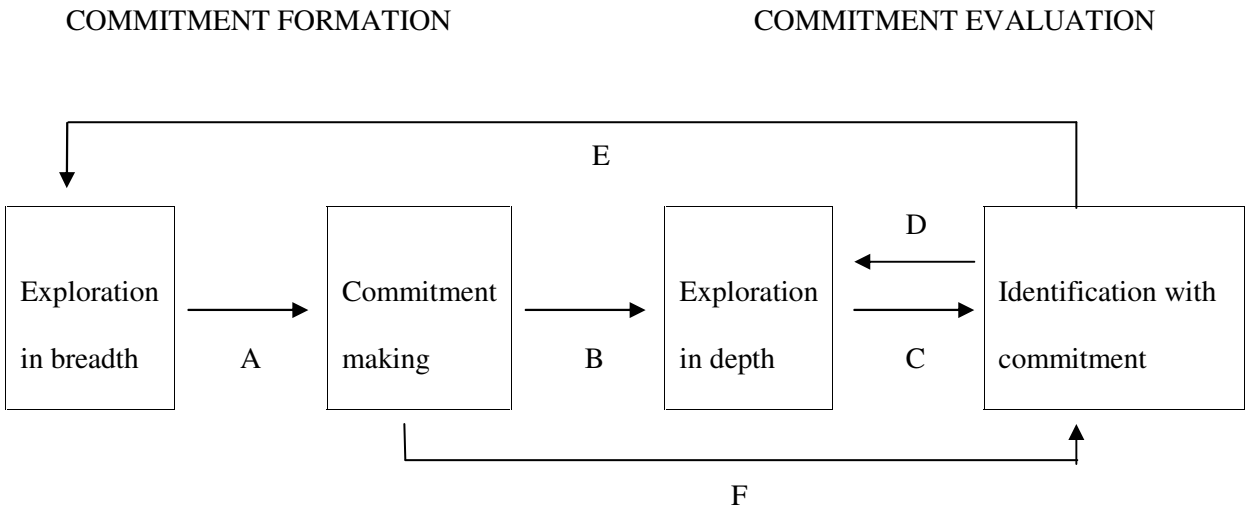

Figure 9.1. Identity developmental model put forward throughout the present dissertation.

A critical characteristic of this model is its iterative or reciprocal nature within and across both identity cycles (Bosma & Kunnen, 2001; Grotevant, 1987; Kerpelman et al., 1997). This appears to be a key feature of identity development in emerging adulthood, which is characterized by continuous explorations and re-evaluations (Arnett, 2004). Although the model might seem rather unidirectional (i.e., the sequence A – B - C), we cannot emphasize enough that this is not the case. For instance, exploration in depth and identification with commitment probably interact in such a way that exploration in depth not only influences identification with commitment, but that temporary identifications with current commitments also influence the need for a prolonged exploration in depth of those commitments (referred to as Path D in Figure 9.1). Further, although – in line with Marcia's (1966) model - exploration in breadth is theorized to act as an antecedent to commitment making, it can be activated at any time during the identity formation process. Adolescents probably reconsider various alternatives when they find the current commitments to be unsatisfying or, conversely, when they identify themselves with their commitments, the need to explore other alternatives diminishes (referred to as Path E in Figure 9.1), as demonstrated in Chapter 5. These reciprocal or feedback loops probably are as central to this model as the developmental order presented earlier and they probably constitute some of the core mechanisms of identity development in the long run. Chapter 8 indeed demonstrated that most emerging adults in the college context evidence some simultaneous identity work on all four identity dimensions.

Different Paths of Development

Although the model is believed to hold for many emerging adults, not everybody follows this same sequence; certain steps are left out in some cases and, for some adolescents, certain steps are not that important. The first example refers to the attainment of a foreclosed identity. Extensive consideration of various identity alternatives before one makes a choice represents an ideal case scenario because that would allow one to evaluate various options before committing to one. Foreclosed adolescents, however, strongly uphold the values and choices provided by significant others, such as parents or society as a whole. Consequently, exploration in breadth prior to making commitments is probably largely ignored by these adolescents (i.e., Path A in Figure 9.1 is almost absent).

The second example considers the subgroup of adolescents who are strongly attuned to their inner wishes and needs. Building on Waterman's (1990, 1993; Waterman et al., 2003) eudaimonistic identity theory and on Deci and Ryan's (1985) self-determination theory (SDT), these individuals probably score high on personal expressiveness, defined as engagement in self-defining activities and commitments that draw upon one's fundamental purposes in life. Personal expressiveness thus refers to the degree to which a person's sense of identity corresponds to her or his unique potentialities (Waterman, 1990). Personal expressiveness also serves as an index of intrinsic motivation (Deci & Ryan, 1985, 2002; Waterman et al., 2003), in that individuals experience a special fit or meshing with their activities or commitments, a feeling of completeness or fulfillment, and an unusually intense involvement. When the alternatives considered are consistent with one's unique and innate potentials, commitment making might constitute a path to self-discovery, that is, of selecting those commitments that reflect one's unique and best potentialities (Schwartz, 2002b; Schwartz et al., 2000). When individuals experience their commitments as personally expressive, they may feel less inclined to proceed to an in-depth social-cognitive evaluation of their commitments (i.e., Paths B and C in Figure 9.1 are less strong in these adolescents). The congruence between such commitments and the person's unique potentials may decrease the perceived necessity of such re-evaluations and a high identification with commitment follows rather automatically after the making of commitments (i.e., path F in Figure 9.1). The path from commitment making to exploration in depth, however, is hypothesized to remain significant for most individuals, suggesting that some evaluation may occur regardless of whether one's commitments are perceived as personally expressive. Further, it remains to be investigated whether - in line with Schwartz (2002b) – exploration in depth also can serve the discovery of who one really is. Further investigation is needed in order to verify all of the paths proposed in Figure 9.1.

Aim 2: External Validation and Correlates of This new Theoretical Framework

Chapters 3 and 4 demonstrated the distinctiveness of the four identity dimensions by relating them to parenting, personality, and adjustment at one point in time (i.e., Wave 1 of L-TIDES). We will summarize the most important results because they provide a stepping stone toward the next section on longitudinal determinants and consequences of identity formation.

Commitment Making and Identification With Commitment

In general, both commitment making and identification with commitment were accompanied by positive intra-individual and contextual characteristics. The similarities in terms of the concurrent associations with the hypothesized antecedents and consequences were striking across the different measurement waves of the L-TIDES project. In terms of the personality correlates, both identity dimensions were negatively correlated with neuroticism and positively correlated with extraversion, agreeableness, and conscientiousness. Both commitment making and identification with commitment were unrelated to openness to experience. In terms of the parenting correlates, both identity dimensions were positively correlated with a composite score of supportive parenting (i.e., high scores on responsiveness and autonomy support, and low scores on psychological control). Further, both identity dimensions were negatively correlated with psychological control. Finally, in terms of the adjustment correlates, both dimensions were positively correlated with self-esteem and negatively correlated with depressive symptoms. Further, the correlations with other adjustment indices (such as substance use and academic adjustment) were also very similar for both identity dimensions.

Previously, we noted that both commitment dimensions were highly interrelated. Consequently, we conducted additional analyses on the data of Wave 1 of L-TIDES to control for the interrelationship between both commitment dimensions - partially inspired by the analytical strategy used in Chapter 3 that aimed at identifying the unique correlates of each identity dimension. The pattern of personality and parenting correlates remained virtually unchanged. These findings suggest that both commitment dimensions possibly stem from or benefit from rather similar parenting or personality characteristics. The pattern of the adjustment correlates of both identity dimensions, however, did change substantially. In line with Chapter 3, especially identification with commitment was significantly related to various adjustment indices. The initial significant correlations between commitment making and the various adjustment indices turned out marginally significant or non-significant when controlling for the relation between commitment making and identification with commitment.

This indicates that the adaptive outcomes associated with making a commitment are largely carried by the extent to which one identifies with the commitment. Note that all associations up till now apply to concurrent correlations.

These findings can be linked to the identity status literature. The statuses characterized by high commitment are accompanied by positive adjustment and the absence of malignant symptoms. On the other hand, the statuses characterized by the absence of commitments are not accompanied by optimal adjustment, but by several possibly debilitating symptoms. For example, individuals in foreclosure and especially achievement displayed the highest self-esteem (Marcia & Friedman, 1970; Prager, 1982; Read, Adams, & Dobson, 1984; Schenkel & Marcia, 1972). Concerning internalizing problem behaviors, individuals in diffusion and especially moratorium reported the most anxiety and depression (Marcia, 1967; Marcia & Friedman, 1970; Podd, Marcia, & Rubin, 1970). Our findings now seem to suggest that these associations might be a function of the certainty and confidence accompanying firm commitments rather than the commitments in and by themselves.

Exploration in Breadth and Exploration in Depth

The intra-individual and contextual correlates of exploration in breadth and exploration in depth proved to be far more divergent than the correlates of both commitment dimensions were. Note that additional analyses in which we explicitly controlled for the interrelationship between both exploration analyses yielded similar results as the ones reported earlier. The latter finding should come as no surprise because the correlation between both exploration dimensions was only moderate and less strong than the correlation between both commitment dimensions. In terms of the personality correlates across the different measurement waves of the L-TIDES project, exploration in breadth was consistently positively related to both neuroticism and openness and negatively to both extraversion and agreeableness. In contrast, exploration in depth was positively related to extraversion and conscientiousness. In terms of the parenting correlates, exploration in breadth was negatively related to concurrent supportive parenting, whereas exploration in depth was positively related to this parenting measure. Finally, exploration in breadth was positively related to psychological control. Exploration in depth, however, was unrelated to psychological control. These findings suggest that the two exploration dimensions – in contrast to the commitment dimensions - do not stem from or benefit from rather similar parenting or personality characteristics.

For instance, it is expected that psychological control is associated with higher levels of exploration in breadth, because emerging adults who experience high levels of psychological

control are known to be anxious and to strongly doubt their actions (Soenens, Vansteenkiste, Luyten, et al., 2005). The sense of indecisiveness that most likely results from such characteristics may lead to a continuous search for alternative options, even to the extent that emerging adults get trapped into a ruminative process of exploring new identity alternatives (Germeijs & De Boeck, 2003). This does not imply, however, that psychological control will also relate positively to exploration in depth. The latter dimension pertains to a more in-depth exploration of one's commitments once they are established. This hypothesis is consistent with recent evidence showing that although psychologically controlling parenting may lead to some behavioral engagement (such as exploring identity alternatives), this engagement is unlikely to result in adaptive, long-term outcomes (Assor, Roth, & Deci, 2004; Vansteenkiste, Simons, Soenens, & Lens, 2004). In other words, adolescents might feel pressured to start exploring identity issues but this exploration most likely involves a superficial type of exploration in breadth.

Finally, in terms of the adjustment correlates, the pattern of the concurrent associations indicated that especially exploration in breadth had a profound impact on various indices of adjustment. At the different waves of the L-TIDES project, exploration in breadth was positively related to depressive symptoms and negatively related to self-esteem. Exploration in depth, on the other hand, was largely unrelated to these adjustment indices. Future research should determine the mechanisms that could explain these concurrent associations with adjustment, such as the parameters of the exploration process and possible moderators.

Parameters of the exploration process. We hypothesize, for instance, that the timing, duration, and/or the intensity of both types of exploration should be taken into account in future research. With respect to exploration in breadth, it will be argued later on in the present chapter that a continued exploration of different identity alternatives throughout emerging adulthood might signal a problematic or ruminative approach towards identity issues when not accompanied by a simultaneous formation of and increase in commitments (Trapnell & Campbell, 1999). An introspection that is too intense might paralyze rather than activate individuals (Bosma & Gerlsma, 2003).

Moderators. The reasons why one explores should also be taken into account in future research. In line with SDT (Ryan & Deci, 2000), we hypothesize that the motivations why one engages in identity work could moderate the link between identity and adjustment. SDT posits that autonomous motivation involves acting with a sense of volition because one finds the activity enjoyable or personally meaningful. Controlled motivation, on the other hand, pertains to feeling pressured to perform a particular behavior because it involves engaging in an activity

to meet an external demand or to live up to a rigid internal standard. Multiple research efforts demonstrated that autonomous (as opposed to controlled) motivation leads to beneficial outcomes in terms of functioning and well-being (Vansteenkiste, 2005). It could be expected that exploration carried out under conditions of autonomous motivation would yield more optimal effects that when carried out under conditions of controlled motivation.

Aim 3: Determinants and Consequences From a Longitudinal Perspective

The central goal of Chapters 6 to 8 was to re-analyze the antecedents and consequences of identity formation from a longitudinal perspective. The theoretical model displayed in Figure 1.2 of Chapter 1 needs some important modifications after all empirical results of the present dissertation have been laid out. Hence, Figure 9.2 gives a graphical representation of the modified theoretical model that includes the hypothesized antecedents and consequences of identity formation. Note that the unidirectional arrows displayed in Figure 1.2 of Chapter 1 are replaced by double-headed arrows in order to comply with the results provided by the different longitudinal analyses (Grotevant, 1987). Careful testing of two competing (sets of) hypotheses, that correspond with unidirectional and bidirectional influences respectively, provided support for the latter set for all three major links represented in Figure 9.1. These associations, as substantiated by reciprocal processes (as identified through the cross-lagged methodology) and common developmental pathways (as identified through the LGCM analyses), refer to the personality – identity, parenting – identity, and identity – adjustment link, respectively.

Personality - Identity

Hypotheses. With respect to the developmental relationships between the Big Five and the four identity dimensions, two general and somewhat competing hypotheses can be put forward. First, it can be hypothesized that personality mainly influences identity and not vice versa. The theoretical distinction between core and surface personality characteristics, as made by Asendorpf and van Aken (2003), further validates this viewpoint. Whereas the Big Five are viewed as core personality traits (McCrae & Costa, 1990), identity formation presumably consists of more malleable individual characteristics. Hence, personality is more likely to influence identity than vice versa. Most cross-sectional research on the link between personality and identity formation to date is conceptualized from this perspective (as represented in Figure 1.2 of Chapter 1).

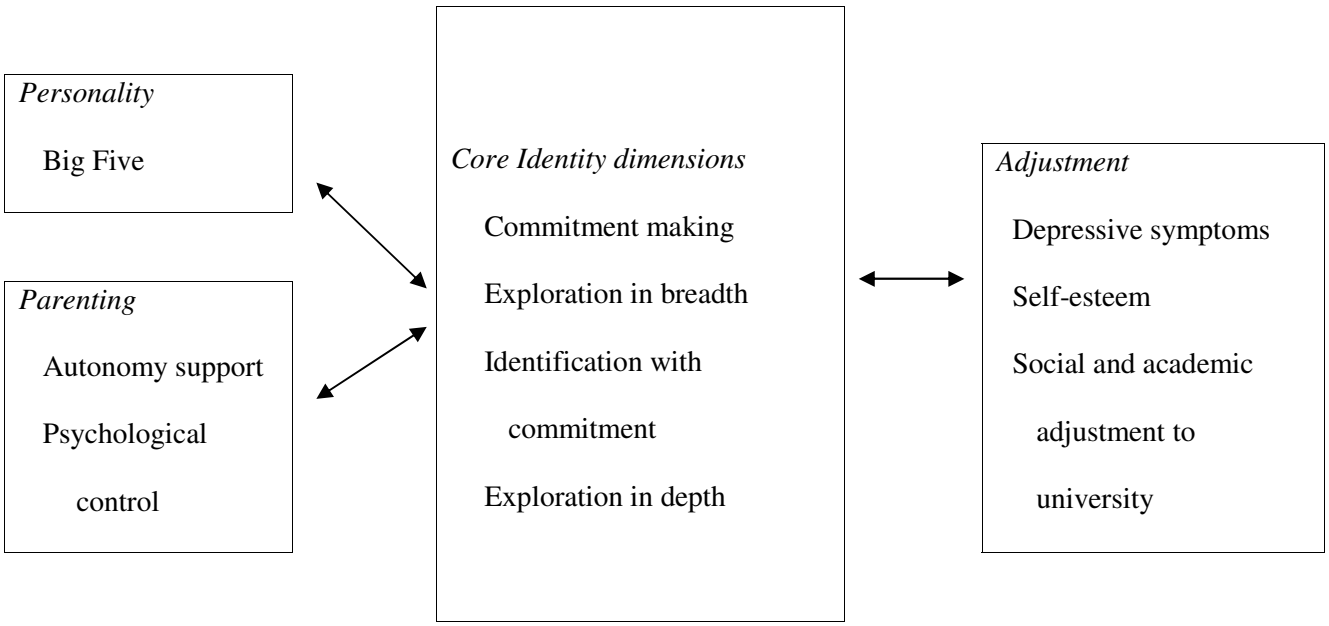

Figure 9.2. Modified theoretical model.

Second, as explained in Chapter 6, it could be hypothesized that identity and personality develop as part of a system with both mechanisms influencing and reinforcing each other (Caspi & Roberts, 1999; Neyer & Asendorpf, 2001). Identity formation could act as a motor for personality changes in the transition to adulthood (Pals, 1999; Robins et al., 2005; Srivastava et al., 2003). The significant amount of self-reflection that identity resolution requires, might in turn promote insight into one's personality and initiate change (Robins et al., 2005). The latter effect illustrates the importance of the process of developing and maintaining an identity towards personality development (Caspi, Roberts, & Shiner, 2005).

Results. The findings of Chapter 6 clearly confirmed the second hypothesis. Reciprocal influences and common developmental pathways between the identity dimensions and the Big Five testified to the existence of an interdependent personality-identity system, with mainly neuroticism, openness to experience, and conscientiousness influencing or being influenced by the different identity dimensions. Neuroticism was connected in a developmental interplay (both at the level of the time-specific measures and underlying developmental trajectories) with both commitment- and exploration dimensions. Further in line with expectations, openness to experience was related especially to the exploration-dimensions, whereas conscientiousness was related especially to both commitment-dimensions.

Earlier in this chapter, we tentatively advanced the idea that both commitment dimensions probably benefit or stem from similar personality traits. The combination of the various longitudinal findings confirmed this hypothesis: both conscientiousness and neuroticism were associated most consistently with the two commitment dimensions. Similarly, we stated

previously that both exploration dimensions probably do not stem from or benefit from rather similar personality characteristics at the group level. Again, the combination of the various longitudinal findings confirmed this hypothesis. Whereas openness was associated most consistently with exploration in breadth, conscientiousness was associated most consistently with exploration in depth.

Future research should focus more on the lower-order traits subsumed under the Big Five to provide more insight into these relationships (Caspi et al., 2005). For instance, with respect to neuroticism, a distinction is made between a cluster of fearful distress (including characteristics such as anxiety, sadness, insecurity, and guilt) and a cluster of irritable distress (including tendencies toward outer-directed hostility, anger, jealousy, and irritation). This distinction is valuable for developmental research because both are likely to follow different developmental paths and predict different outcomes (Caspi et al., 2005). If one could identify which of these two subfactors would be most influential towards or influenced by identity dimensions such as commitment or exploration, this would have important implications for intervention. Individuals scoring high on fearful distress and individuals scoring high on irritable distress probably benefit from different intervention strategies, much in the same vein as individuals suffering from internalizing problems benefit from different intervention practices than do individuals suffering from externalizing problems.

Similarly, with respect to conscientiousness, an ongoing debate involves the question of how the high end of this trait should be conceptualized (Caspi et al., 2005). Highly conscientious people are generally responsible, careful, persistent, orderly, and planful which could explain the consistency in both our cross-sectional and longitudinal findings (i.e., conscientiousness is positively associated with both commitment dimensions). However, being high on conscientiousness also has been seen as a willingness to follow authority and to conform to group norms. Hence, Caspi et al. (2005) raised the question whether it is possible to be too conscientious, resulting in some sort of closed-mindedness. Translated to our findings, one could ask if persons high on conscientiousness adopt a more information-oriented, explorative approach (Berzonsky, 1990) or a more foreclosed approach to identity issues. Duriez et al. (2004) found conscientiousness to be unrelated to an information-oriented approach towards identity issues. However, a normative, foreclosed approach towards identity issues was positively related with conscientiousness. Future research should further look into the link between conscientiousness, commitment, and exploration.

Parenting - Identity

Hypotheses. Current research on family socialization and parenting witnesses a strong interest in the construct of parental psychological control, a parenting dimension which is highly relevant to the process of identity formation because it intrudes upon or impedes the adolescent's search for autonomy. "[Psychological control] is characterized by a type of interpersonal interaction in which the parent's psychological status and relational position to the child is maintained and defended at the expense and violation of the child's development of self" (Barber & Harmon, 2002, p. 6). As noted in Chapter 7, psychological control creates a climate in which dysfunctional or maladaptive identity processes will be initiated or exacerbated, and, conversely, in which self-initiation and self-governing will be impaired (Barber, Bean, & Erickson, 2002).

With respect to the developmental relationships between this parenting dimension and identity, the same two general hypotheses advanced for the Big Five – identity link can be stated. First, it is generally accepted that family functioning influences the identity formation process of adolescents and emerging adults (Adams & Jones, 1983; Adams, Ryan, & Keating, 2000; Grotevant & Cooper, 1986). Adolescents who experience positive interactions with their parents and who perceive their parents as being supportive tend to develop an integrated set of commitments with which they can identify. Conversely, a more fragmented sense of identity or self can emerge when parents are not sensitive to or attuned to the needs of their adolescent children. Recent studies indeed confirmed that psychological control exacerbates, rather than merely accompanies, indices of psychosocial functioning (Aunola & Nurmi, 2004; Barber et al., 2005; Conger, Conger, & Scaramella, 1997).

Second, in line with transactional models of socialization (Grotevant, 1987; Laird & Pettit, 2001; Magnusson, 1988; Stattin & Kerr, 2000), we hypothesized that reciprocal relations between parenting and emerging adult identity formation would emerge. Grotevant (1987) already stated that the contexts of development – those aspects of one's society, family, peers, school, and work environments in which the individual's sense of identity is created – affect the identity formation process. However, these contexts in turn are also shaped by individual's choices within these settings (Adams & Marshall, 1996).

Results. Again, the findings of Chapter 7 provided support for the second hypothesis. First, emerging adults who perceived their parents as intrusively controlling appeared to have serious difficulties in establishing clear and committed choices in important life areas. Moreover, to the extent that they did manage to make commitments, they were unlikely to fully identify with them or to experience a sense of certainty and satisfaction concerning these

commitments because they probably become detached from their inner feelings and needs (Bosma & Gerlsma, 2003). Conversely, parents who are not responsive to their children's needs and who use psychologically controlling techniques frequently frustrate their children's need for autonomy and volitional functioning, and, consequently, probably hinder the formation of strong and personally endorsed commitments. Further, these findings are in line with past studies showing that children of psychologically controlling parents experience high levels of fear of failure and tend to be overly concerned with making mistakes and failing to live up to one's personal standards (Elliot & Thrash, 2004; Soenens, Vansteenkiste, Luyten, et al., 2005). Such a perfectionist and anxious orientation most likely results in the procrastination of identity-relevant choices and in a relentless negative evaluation of the few commitments that one has already made. In sum, it is hypothesized that two different but interrelated mechanisms of action in the link between psychological control and commitment could be (a) the frustration of one's sense of autonomy (viewed as living in accordance with one's true inner values and needs) and (b) the installment of a maladaptive perfectionistic orientation that prolongs the decision process needed to make firm identity commitments. Further research should investigate the viability of these hypothesized mediators.

Second, the reverse path – from identity to parenting – became apparent in the longitudinal link between exploration in breadth and psychological control. It seems that psychological control also emerges as a reaction to a continuous search for identity alternatives because parents might want to pressure their offspring to arrive at firm identity commitments, especially when these children are already in their late teens or early twenties. Parents may feel anxious or disappointed about their emerging adult children's continuous search for new identity options because this search could forestall a choice for a clear direction in life. As a consequence, parents may attempt to pressure their children to make a certain commitment which reflects the parents' own interests and values rather than their children's interests and values, thereby using intrusive methods such as guilt-induction and conditional approval. These parenting strategies, however, are accompanied by certain outcomes that are probably contrary to the intentions of the parents, that is, a further forestallment of identity-related commitments due to the mechanisms explained earlier.

Previously in this chapter, we stated tentatively that both commitment dimensions may benefit or stem from similar parenting climates at the group level. Again, the combination of the various longitudinal findings on parenting and identity confirmed this hypothesis: psychological control hindered both commitment dimensions probably in a similar fashion. Similarly, we stated that both exploration dimensions probably do not stem from or benefit

from rather similar parenting climates. The findings from the longitudinal analyses presented herein, however, failed to provide unequivocal evidence for this hypothesis.

Apparently, parents of emerging adults continue to play an important role in the daily lives of their children, although approximately 80% of our sample only saw their parents during the weekends. Our results concurred with Arnett's (2004) thinking by indicating how important it is for parents to view their emerging adult children as equals, respectful of their inner wishes, standards, and needs (Allen, Hauser, Bell, & O'Connor, 1994; Chirkov & Ryan, 2001; Dekovic & Buist, 2004; Kurdek, Fine, & Sinclair, 1995; O'Connor, Allen, Bell, & Hauser, 1996; Soenens & Vansteenkiste, 2005; Steinberg, Elmen, & Mounts, 1989; Vansteenkiste, Simons, Lens, Soenens, & Matos, 2005). Parents who behave in such a way are described as being high on autonomy support as defined within SDT (Deci & Ryan, 1985).

Luyckx, Soenens, Goossens, and Vansteenkiste (2005) relied on a longitudinal mediation model (Baron & Kenny, 1986; Cole & Maxwell, 2003; Holmbeck, 1997; Rose, Holmbeck, Coackley, & Franks, 2004) and focused on identity formation as a mediator between parenting and adjustment in the academic domain. They demonstrated that autonomy-supportive parents who use little psychologically controlling techniques probably helped their children to be aware of and to behave according to self-initiated and authentic goals (Grolnick, 2002; Vansteenkiste, 2005). As a consequence, these children were more likely to engage in activities and make identity-related choices in the academic domain in accordance with their inner value system. Because their choices were based upon and fit in their inner value system, these adolescents were more likely to feel confident and certain about them, which, in turn, facilitated academic adjustment. Soenens, Vansteenkiste, Lens, et al. (2005) demonstrated that such a parenting dimension of autonomy support needs to be distinguished from the parenting dimension of promoting independence. In the empirical and theoretical literature to date, however, both conceptualizations are frequently used interchangeably (Gray & Steinberg, 1999; Grolnick, 2002; Silk, Morris, Kanaya, & Steinberg, 2003).

The notion of promoting independence stems from the theory of separation-individuation (Blos, 1979; Goossens, in press; Kroger, 1985; Kroger & Haslett, 1988; Levy-Warren, 1999). This theory states that adolescents must gradually distance themselves physically and emotionally from their parents (i.e., separation) and increasingly take responsibility for themselves without strongly depending or relying on their parents (i.e., individuation or independence). Within separation-individuation theory, it is claimed that these processes of separation and individuation are a sign of maturation and would result in more adaptive psychosocial and developmental outcomes (Blos, 1979). Empirical research, however, has

failed to provide consistent evidence for such claims (Goossens, in press; Soenens, Vansteenkiste, Lens, et al., 2005). Separating oneself from parental influences and developing independent functioning carries emotional costs and may sometimes even be at odds with a positive parent-child relationship (Ryan & Lynch, 1989).

Results obtained in L-TIDES provided some support for the latter hypothesis. We derived a composite score of emotional independence (i.e., the freedom from a need for approval, closeness, and relational support in relation to parents) and functional independence (i.e., the ability to manage and direct one's practical and personal affairs without parental help) from parents - as measured with two subscales of the Psychological Separation Inventory; PSI, Hoffmann, 1984). This composite score could be viewed as a primary outcome of parental promotion of independence (Beyers, 2001). Multivariate LGCM analyses for the first 5 waves of L-TIDES (df = 18; χ^2 = 62.05 (p < .001); RMSEA = .08; CFI = .98; SRMR = .02) demonstrated that the slope of independence was positively related to the slope of depressive symptoms (r = .45, p <.01) and negatively to the slope of self-esteem (r = -.23, p <.05). Growing more independent from parents was accompanied by a simultaneous increase in depressive symptoms and a simultaneous decrease in self-esteem.

Further, it is generally assumed that growing independent from parents goes hand in hand with progressive identity formation in late adolescence (Levy-Warren, 1999). In Chapter 3, it was hypothesized that one apparently has to find a balance between distancing oneself from parents and becoming close again, depending on one's developmental phase. The making of commitments would be promoted by being close to parents, whereas exploring various alternatives would benefit from some distance between parents and their adolescent children (Kaplan & O'Connor, 1993). This idea could be only partially replicated longitudinally. Multivariate LGCM-analyses (df = 18; χ^2 = 62.05 (p < .001); RMSEA = .08; CFI = .98; SRMR = .02) indicated that the slopes of independence and commitment making were negatively related (r = -.21, p < .05), meaning that growing more independent from parents is accompanied by less strong identity commitments. However, the slopes of independence and exploration in breadth were not significantly correlated (r = -.02, p > .05), a developmental finding that disconfirms the second part of the hypothesis stated earlier.

In sum, separation-individuation and identity formation need not to go hand in hand. This is in line with contemporary critiques on separation-individuation theory. In essence, these critiques state that healthy development is possible without separation from parents (Kroger, 2005). Positive autonomy development needs to be defined in terms of processes of self-

governance that do not preclude close and interdependent relationships with parents (Grotevant & Cooper, 1986; Hill & Holmbeck, 1986).

Identity – Adjustment

Hypotheses. With respect to the developmental relationships between adjustment (i.e., self-esteem and depression) and the four identity dimensions, the same two sets of general hypotheses were advanced. First, many identity researchers – although mainly based upon cross-sectional research - argued for the existence of a unidirectional causal path going from identity formation to adjustment. Adjustment was generally viewed as an outcome of identity formation. Second, it was hypothesized, in line with identity theorists such as Grotevant (1987) and Schwartz (2001), that reciprocal effects would occur (cf. Vleioras & Bosma, 2005). Schwartz (2001), for instance, stated that satisfaction and well-being appear to be associated with commitment and, as stated previously, are likely to inhibit exploration to some degree.

Results. The findings of Chapter 8 provided support for the second general hypothesis. First, commitment making and identification with commitment appeared to predict subsequent levels of both well-being and distress in a rather similar vein. Apparently, the conclusions reported earlier on the concurrent link between both commitment dimensions and adjustment are partially confirmed in the longitudinal analyses but need some modification as well. Whereas commitment making was rather unrelated to or only marginally related to adjustment at the concurrent level when controlling for the interrelation between both commitment dimensions, commitment making did impact on depressive symptoms in the longer term, as demonstrated in the longitudinal analyses.

The broad exploration of identity alternatives gave rise to increasing depressive symptoms. When students were still exploring various identity alternatives to a considerable degree in the first year of college, a detrimental influence on their adjustment was observed. Exploration in depth also constituted a predictor of distress – in contrast to the cross-sectional findings -, through its negative impact on subsequent self-esteem between the second and third year of university. Apparently, continuing exploration in depth during emerging adulthood might also be indicative for some individuals of an underlying uncertainty adversely affecting mental health. These findings are partially in line with the suggestion stated earlier that the timing, duration, and/or the intensity of exploration in depth should be taken more explicitly into account in future research. Of course, important interindividual differences are expected to emerge. What constitutes an appropriate amount or timing of exploration in depth for one adolescent may not necessarily do so for another adolescent. This will probably also depend on

the developmental position of the individual on the other three identity dimensions, as will be demonstrated later on in the section on developmental trajectory classes.

Second, a lone adjustment-identity path – from self-esteem to identification with commitment – suggested an interesting reciprocal loop possibly at work in the identity-adjustment link. High commitment making appears to promote subsequent self-esteem, and, in turn, high self-esteem appears to promote subsequent identification with commitment. The latter, in turn, again promotes subsequent self-esteem. This process may suggest a mechanism by which individuals, particularly those who adhere strongly to their current commitments, maintain high self-worth and avoid considering other alternatives (Archer & Waterman, 1990; Marcia, 1980). Later on in this chapter – in the section on the developmental trajectory classes – we will describe an identity development class labeled the consolidators, consisting of individuals who are very strongly focused on their current commitments without considering alternatives. They demonstrated the most optimal adjustment as compared to the other identity development classes in the sample. This mutually reinforcing commitment – self-esteem pathway might be an important mechanism of action that could account for the high levels of well-being in this consolidator class.

Aim 4: General Developmental Trends in Identity

This study (L-TIDES) provided unique descriptive information on identity development at the average level in the university context. Figure 9.3 presents a graphical overview of these observed developmental trends.

Apparently, both dimensions of the commitment evaluation cycle (i.e., exploration in depth and identification with commitment) are more activated at each time point in comparison to both dimensions of the commitment formation cycle (i.e., exploration in breadth and identification with commitment). The age period under study appears to be a time in life in which the evaluation of commitments comes to the fore, although the formation of commitments continues to be important (Arnett, 2000, 2004). Note, however, that the mean differences among the four identity dimensions are small and that there is substantial variability for each dimension at each measurement time.

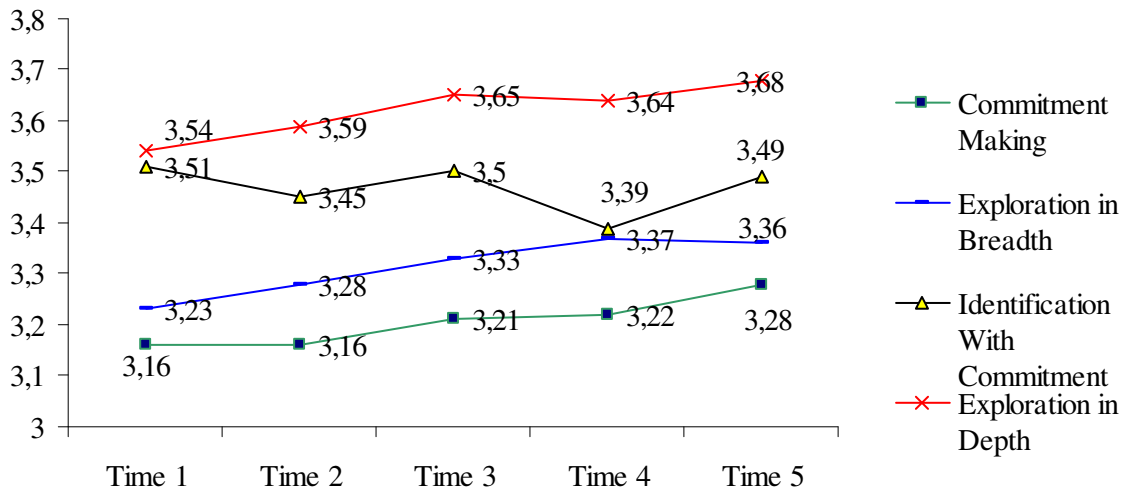

Figure 9.3. Observed developmental trends in identity for the first five measurement waves of L-TIDES.

All identity dimensions, except for identification with commitment, tended to increase with age. These findings appear to be in line with previous research focusing on trends in these identity dimensions (van Hoof, 1999a). These mainly cross-sectional studies reported that, in general, all four identity dimensions – as assessed through identity status interviews or the U-GIDS - appeared to remain stable or increase with age, depending on the study. For exploration in breadth, however, a flattening of the increasing trend seems to emerge at the fifth measurement wave in our study. The degree to which this trend perseveres or even becomes stronger (i.e., a decrease instead of an increase in strength of exploration in breadth) needs to be followed-up. The latter could mean that with increasing commitment making, the motivation to engage in exploration in breadth might gradually decrease (Grotevant, 1987). Previous longitudinal research on the identity statuses in emerging adults and adults (Fadjukoff et al., 2005; Pulkkinen & Kokko, 2000) indicated that both identity diffusion and moratorium tended to decrease with age whereas both achievement and foreclosure tended to increase with age. These observations lend some credit to our observation that exploration in breadth might decrease at some point in emerging adulthood.

The trajectory of identification with commitment seemed to go up and down. In Chapter 5, it was hypothesized that these fluctuations in identification with commitment could very well be an important mechanism inherent in commitment evaluation. It could be precisely this fluctuation that constitutes the outcome of a continuous, iterative evaluative process. In turn, these fluctuations concerning the current commitments could instigate ongoing evaluative work. Future research should also focus on other factors (both intra-individual and contextual)

that could account for these fluctuations in identification with commitment. For instance, certain (life) events experienced on the way (e.g., finding or breaking up with one's girlfriend/boyfriend or getting one's semester grades) could account for these short-term increases or decreases in identification with commitment and hence, could function as important markers of identity development (Grotevant, 1987). In a similar vein, Kalakoski and Nurmi (1998) and van Hoof (1999a) made a distinction between the maturation or stage hypothesis and the transition or life-events hypothesis in looking at identity development. Whereas the first hypothesis assumes that developmental identity changes take place with age, the second hypothesis assumes that certain changes outside the individual (e.g., life events or environmental changes) have the capacity to induce changes.

Aim 5: Developmental Trajectory Classes

A focus on distinct developmental trajectory classes instead of on general developmental trends requires a substantial shift in approaching the data at hand. In the latter approach (i.e., the variable-centered approach) one is interested in uncovering general trends and interrelationships valid for the whole sample, whereas in the former approach (i.e., person-centered approach) one wants to differentiate among meaningful subgroups within the sample to investigate whether these subgroups behave differently on a number of correlates, antecedents, or outcomes (Bates, 2000; Magnusson, 1998; Nagin, 2005). This approach is of great theoretical relevance due to the heterogeneity of identity formation in the transition to adulthood (Arnett, 2004). The clinical benefits of such an approach are also obvious. It allows researchers to identify extreme groups of individuals that might be possible candidates for intervention.

Typologies of Identity Formation

Several authors have identified different types of identity formation in adolescence and adulthood. Table 9.1 gives an overview of these different typologies. All identity types or classes in the same column correspond to one another. All authors distinguished among 4 identity statuses, types, or classes, except Côté and Levine (2002) who distinguished among 5 identity types on a theoretical basis. In Chapter 8, we also identified 4 longitudinal identity classes, 3 of which corresponded to identity types described in the literature. We also identified a new longitudinal identity class, that is, the Consolidators.

Josselson's (1996) identity classes and Marcia's (1966) identity statuses were described earlier in this dissertation. Chapters 1 and 8 described how identity statuses are commonly

derived. Josselson (1996) relied on qualitative data and traced the identity development of a group of 30 women over a 22-year period beginning at the end of their college years. She used a slightly modified version of Marcia's identity status interview to classify individuals in one of four groups. Once assigned to a status group after the initial college interviews, however, individuals were not reassigned to other status groups based on later interviews; original group membership was retained. Apparently, the classification in identity categories did not capitalize fully on the developmental nature of the data, as is routinely the case in longitudinal research on the identity statuses.

Table 9.1

Typologies of identity formation in adolescence and adulthood

Authors	Different classes or types				
Marcia (1966)	Achievement	Moratorium	Foreclosure	Diffusion	--
Josselson (1996)	Pathmakers	Searchers	Guardians	Drifters	--
Côté & Levine (2002)	Resolvers	Searchers	Guardians	Drifters/Refusers	--
Helson & Srivastava (2001)	Achievers	Seekers	Conservers	Depleted	--
Luyckx, Schwartz, Goossens, & Soenens (2005)	Pathmakers	Searchers	Guardians	--	Consolidators

Other authors also delineated similar trajectory classes in late adolescence and adulthood. In the social-psychological literature, Côté and Levine (2002) – inspired by Josselson's (1996) typology - made a theoretical distinction between 5 typical identity formation strategies in late-modern societies, that is, refusers, drifters, searchers, guardians, and resolvers. Côté and Levine proposed this typology to demonstrate that there currently are a greater number of ways in which adult identities are formed, most of which involve some degree of individualization.

The resolvers are actively engaged in the process of forming an identity, fully capitalizing on the opportunities provided in late-modern societies in general or the college context in particular. They interact with the surrounding world (i.e., they are open to various opportunities

and explore different possibilities) in order to reach psychosocial and emotional maturity. Such a growth orientation is probably fed by a motivational desire to reach one's inner potentials, a desire which can be stimulated by understanding, empathic, and supportive parents. Searchers are often driven by unrealistically high standards and criteria of functioning, making them unable to form a steady set of commitments. They seem to be locked in a perpetual journey for which there is no end because they are unable to find perfection within themselves or in the community (Côté & Levine, 2002). This may create a sense of despair because searchers are driven by a sense of dissatisfaction. In contrast, the guardians have thoroughly internalized the values of their parents or of the society, providing them with a set of determined resources to move into adulthood. However, the rather rigid nature of this process, insensitive to change, comes with several disadvantages. These individuals can neglect to undergo certain developmental experiences that would allow intellectual and/or emotional growth. Further, this rigid approach can result in stereotyped relationships.

Finally, both the refusers and the drifters appear to handle the demands of adult identity formation rather similarly. They evidence a behavior pattern characterized by poor impulse control, shallow interpersonal relationships, and a lack of steady commitments to an adult community. Both subgroups, however, are distinguished at the level of resources they have at their disposal to handle the identity formation task. Whereas refusers probably have little personal resources (e.g., emotionally and intellectually) at their disposal, drifters have more personal resources at their disposal but they seem unable or uninterested in applying them. Especially refusers were likely to receive little structure, guidance, and encouragement from their social environment in general and their parents in particular as children or adolescents. In sum, Côté and Levine (2002) aimed at introducing a tentative working typology of identity strategies that seem to capture the range of contemporary life-course trajectories but the specific mechanisms and processes hypothesized to be at the heart of these various types require a thorough empirical inquiry.

Helson and Srivastava (2001) made a distinction between four distinct classes of adult development in midlife women, based upon two underlying dimensions, that is, environmental mastery and personal growth (Ryff, 1989). Environmental mastery was defined as the ability to achieve a good fit to the environment, to have a sense of mastery and competence in managing it, and to make effective use of surrounding opportunities. Personal growth was defined as the ability to see the self as growing and expanding in ways that reflect more self-knowledge and effectiveness. Individuals scoring high on both dimensions were labeled achievers. Individuals scoring high on personal growth but low on environmental mastery were labeled seekers.

Individuals scoring high on environmental mastery but low on personal growth were labeled conservers. Finally, individuals scoring low on both dimensions were labeled depleted.

Each of these four groups advanced in certain directions throughout their lives in terms of mental health, intimacy, and identity. Achievers displayed a conscientious, outgoing orientation with a relative absence of neuroticism and they tended to accentuate positive emotions. Further, superior effectiveness and identity integration were found to be predominant in these individuals. Seekers were open for new experiences and searched for new experiences at the cost of anxiety and evidenced the least identity acceptance (or, conversely, the most identity questioning). Further, they experienced the full range of both positive and negative emotions. Conservers were motivated to avoid anxiety, seek security, and were described as readily accepting social norms and values. They scored the lowest on openness to new experiences and were characterized by relatively subdued emotions. Finally, the Depleted seem to lack confidence, psychological resources, and identity integration. Further, they lack a positive mental health pattern.

Identity Developmental Trajectory Classes With Latent Class Growth Analysis

The distinction between these similar classes coming from different angles and theoretical backgrounds emphasizes the need to identify identity classes using an explicit quantitative and developmental or longitudinal approach. To the best of our knowledge, such an analysis has never been reported. Chapter 8 of this dissertation aimed at distinguishing between identity trajectory classes using multivariate statistical techniques that fully capitalize on the developmental nature of the data obtained (i.e., Latent Class Growth Analysis; LGCA, Nagin, 2005). This means that the trajectory classes were empirically defined based upon the longitudinal trends - in terms of initial level and rate of change - of the constituting identity dimensions. As can be seen in Table 9.1, a solution with four distinct longitudinal clusters was retained (i.e., pathmakers, searchers, guardians, and consolidators) that mirrored to a large degree the typologies identified in the literature. These four trajectory classes will be described in terms of their developmental position on the underlying identity dimensions, their associations with adjustment (see Chapter 8), and their associations with parenting dimensions and the Big Five of personality.

Additional multivariate analyses of variance (MANOVAs) were performed, six in all. The first set of three MANOVAs focused on four parenting dimensions (i.e., responsiveness, behavioral control, psychological control, and autonomy support; Soenens et al., 2004) assessed at Times 1, 3, and 5, respectively, as dependent variables and the four identity trajectory classes

as independent or fixed variable. The second set of three MANOVAs focused on the Big Five of personality assessed at Times 1, 3, and 5, respectively, as dependent variables and, again, the four identity trajectory classes as independent variable. Because the MANOVAs yielded very similar results for these three measurement waves, only the results of the first wave will be presented. If substantial differences among the three waves did occur, they will be discussed. Table 9.2 provides an overview of the MANOVAs for both personality and parenting at Wave 1 (Luyckx & Goossens, 2006; Luyckx, Soenens, Goossens, & Beyers, 2006). As described in Chapter 8, these analyses should be interpreted with caution because LCGA focuses on latent group membership or, put differently, the trajectory classes identified have a probabilistic nature (Nagin, 2005). MANOVA, however, assumes that group membership is observed or deterministic. Nonetheless, because of the high classification accuracy of our LGCA-analyses, the MANOVAs were assumed to be highly precise (Shaw et al., 2003; Windle & Wiesner, 2004).

Pathmakers. To summarize, pathmakers displayed a high degree of commitment making and exploration in breadth at the beginning. Both these dimensions of the commitment formation cycle showed strong upward trends during the first two years of college (with exploration in breadth even showing the strongest upward trend as compared to the other identity classes). Clearly, this is a developmental translation of Marcia's (1966) achievement status. With regard to the dimensions of the commitment evaluation cycle, both identification with commitment and exploration in depth were initially high (with the latter dimension being the highest as compared to the other identity classes). Whereas identification with commitment remained at this same high level, exploration in depth further increased. Apparently, this cluster of individuals – much in the same vein as the resolvers (Côté & Levine, 2002) - is very active both in the forming and strengthening of commitments (with exploration in breadth at the heart of this process) and in the evaluation of current commitments (with exploration in depth at the heart of this process).

With regard to adjustment, the pathmakers overlapped almost completely with the trajectory classes of increasing adjustment and optimal adjustment, as described in Chapter 8. Table 9.2 indicated that this group, compared to the other three trajectory classes, scored high on parental responsiveness and autonomy support, as could be expected from the typologies described earlier. The degree of psychological control in the pathmaker class was significantly lower than in the searcher class. With regard to personality characteristics, the pathmakers scored relatively low on neuroticism and relatively high on the other four personality factors. Apparently, these individuals have a wide range of resources or strengths at their disposal (Côté

& Levine, 2002), both at the intra-individual personal level and the micro-contextual level (Adams & Marshall, 1996), to face the identity formation tasks at hand.

Table 9.2

Univariate ANOVA's and Post-hoc Trajectory Class Comparisons for Parenting and the Big Five at Wave 1

	Trajectory classes				*F* (3, 392)	Eta²
Dimension	Searchers	Guardians	Pathmakers	Consolidators		
Parenting						
Responsiveness	3.29[c] (.86)	3.68[b] (.80)	3.80[b] (.84)	4.16[a] (.64)	14.56*	.10
Behavioral	3.17 (.78)	3.14 (.66)	3.12 (.67)	3.03 (.64)	0.54	.00
Psychological control	2.23[a] (.66)	1.94[b] (.61)	1.97[b] (.66)	1.66[c] (.50)	10.47*	.07
Autonomy	3.76[c] (.65)	3.93[bc] (.66)	4.00[ab] (.62)	4.22[a] (.48)	6.92*	.05
Big Five						
Neuroticism	3.21[a] (.63)	2.80[b] (.62)	2.93[b] (.62)	2.76[b] (.57)	9.31*	.07
Extraversion	3.39[b] (.57)	3.64[a] (.46)	3.64[a] (.47)	3.77[a] (.40)	8.63*	.06
Openness	3.55[a] (.44)	3.33[b] (.43)	3.60[a] (.39)	3.31[b] (.39)	13.08*	.09
Agreeableness	3.59[c] (.39)	3.74[b] (.38)	3.77[b] (.39)	3.92[a] (.36)	9.28*	.07
Conscientiousness	3.16[c] (.54)	3.38[b] (.40)	3.55[ab] (.43)	3.67[a] (.42)	20.21*	.13

Note. A trajectory class mean is significantly different from another mean if they have different superscripts. A mean without a superscript is not significantly different from any other mean. Standard deviations are in parentheses.

*$p < .001$.

These individuals are probably characterized by a trajectory of developmental individualization (Côté, 2002; Côté & Levine, 2002; Côté & Schwartz, 2002), as opposed to the trajectory of default individualization, both situated at opposite ends of a continuum (Schwartz, 2002b). Developmental individualization represents consciously and deliberately searching for growth opportunities, including identity options that are consistent with one's potentials. The person following this trajectory transacts with the environment in a purposeful way, and may even utilize the lack of structure in late-modern society as an opportunity to form a unique and self-directed sense of identity. Thus, the individual evaluates identity opportunities according to their potential for furthering self-development and for providing personal benefit and advancement (Schwartz, 2002b). These individuals take advantage of social resources and

possibilities in an active manner to form, develop, and revise or evaluate their identity on the way to self-realization (Côté, 2002; Kerpelman et al., 1997; Schwartz, 2002b). This conceptualization of developmental individualization overlaps somewhat with Stephen et al.'s 1992) description of the experiential orientation (in contrast to the instrumental orientation). Rather than accepting societal and/or parental constructions of what is meaningful, the experientially oriented individual needs to discover this for him- or herself in order to construct a coherent and personally authentic worldview.

Guardians. To summarize, guardians initially displayed a moderate degree of commitment making and a rather low level of exploration in breadth. Whereas commitment making showed an upward trend over time, exploration in breadth continued to be rather low, in line with Marcia's (1966) foreclosure status. With regard to the dimensions of the commitment evaluation cycle, identification with commitment was moderate and this continued to be the case. Exploration in depth, on the other hand, was the lowest as compared to the other developmental classes and increased slightly across time. Apparently, these individuals not only are rather closed to new identity options, they also do not think very thoroughly about current commitments as compared to other individuals. This cluster of individuals thus – to some degree – resemble the firm foreclosures identified by Kroger (1995) or the closed foreclosures identified by Archer and Waterman (1990), with both subgroups being rather defensive when it comes to information that could possibly endanger the strength of current commitments.

With regard to adjustment, results are quite similar as for the pathmakers. Practically all guardians overlapped with the trajectory classes of increasing adjustment and optimal adjustment. Apparently, these individuals were characterized by simultaneous well-being, as is routinely reported for foreclosures (e.g., Marcia, 1980). Further, as demonstrated in Table 9.2, they scored very similar as the pathmakers on the parenting dimensions and the Big Five, with one exception. They scored significantly lower on openness than the pathmakers did, in line with Helson and Srivastava's (2001) description of conservers and Côté and Levine's (2002) description of guardians.

Referring to the distinction between developmental individualization and default individualization made earlier, Schwartz, Côté, et al. (2005) found foreclosures to score higher on indices of default individualization than of developmental individualization, in contrast to pathmakers or achievers. Default individualization refers to a life course dictated by circumstance, with little agentic assertion on the part of person (Côté, 2000) - with agency being defined as a sense of responsibility for one's life course, the belief that one is in control of one's decisions and is responsible for their outcomes, and the confidence that one will be

able to overcome obstacles that impede one's progress along the way (Côté & Levine, 2002; Schwartz, Côté, et al., 2005). These individuals ignore or fail to create opportunities for self-improvement and, hence, they are unprepared to address some issues that prepare them to successfully undertake several adult roles. Instead, they select identity options without much effort or consideration, dictated to them by trends and forces in the social environment.

In this respect, Côté and Schwartz (2002) described foreclosure as some sort of developmental arrest because this status is predominantly characterized by the adoption of external norms and values without questioning them. However, the present results demonstrate that – during the first years of emerging adulthood – this state of foreclosure does not come with high costs in terms of adjustment. Further, these individuals could be barely differentiated from pathmakers – being the prototype of developmental individualization – on parenting and personality. Future research should determine whether a foreclosed approach to identity issues indeed is a developmental setback by conducting long-term longitudinal research well into adulthood to see how these individuals cope with demanding adult tasks.

Searchers. To summarize, searchers initially scored the lowest on commitment making and the highest on exploration in breadth. Further, whereas the former dimension remained stable, exploration in breadth further increased over time. With respect to the two dimensions of the commitment evaluation cycle, a rather similar picture emerged. Identification with commitment was the lowest as compared to the other three developmental classes and further decreased across time – the searchers are the only individuals in the sample for which identification with commitment decreased. Exploration in depth, however, was moderate and further increased. Apparently, the identity work undertaken by these individuals not only has to be situated at the level of exploring in breadth various alternatives to guide the formation of commitments (as is routinely assumed in the identity literature), but also at the level of exploring in depth the tentative commitments that are already there.

The in-crisis position of some of these individuals is illustrated when looking at how they overlap with the adjustment trajectory classes. In contrast to the pathmakers and guardians, a substantial proportion overlaps with the stable maladjustment class. Similar findings instigated Côté and Schwartz (2002) to question the developmental adequacy of moratorium as a response to the identity stage, especially when the continued exploration may be more characterological or a rather stable personality feature (Marcia, 1980, 2002). Côté and Schwartz hypothesized that for these individuals, moratorium may have more in common with diffusion than with achievement in terms of agency and decision-making, much in line with how Côté and Levine (2002) viewed their searcher category.

In sum, a substantial portion of the searchers probably are characterized by a trajectory of default individualization which could lead to confusion and dissolution of identity (Stephen et al., 1992). However, it should be emphasized that the majority of the searchers displayed positive well-being in our study. Apparently, intense identity searching without the formation of firm simultaneous commitments can also serve the route to personal growth for some emerging adults (Arnett, 2000; Helson & Srivastava, 2001) and, hence, probably should be viewed in the light of personal development or self-discovery and not of maladaptive rumination or procrastination (Schwartz, 2002b). Helson et al. (1995) made a distinction between integrated searchers and unintegrated searchers with the latter class of individuals experiencing great difficulty in reconciling or integrating needs and goals with their identity, much in the same vein as diffused (or drifting) individuals do. The existence of these distinct subgroups could have important implications for intervention.

With respect to the perceived parenting dimensions, the searchers scored the lowest on parental responsiveness and autonomy support and the highest on psychological control. As stated in Chapter 7, parenting and identity probably influence one another. The lack of empathy and respect for the adolescent's needs and inner wishes, coupled with the use of intrusive and controlling techniques, probably gives rise to an identity approach which is typical for searchers: a continuous search for identity alternatives, interspersed with periods of drifting, without the power to make firm commitments. In other words, these individuals have difficulty to judge whether certain options fit with the inner needs partially due to the self-constraining parental environment that leads to an anxious and avoidant orientation.

With respect to the Big Five, Table 9.2 demonstrates that the searchers are characterized by being high on openness and neuroticism and low on extraversion, agreeableness, and conscientiousness. Clearly, this pattern of scores is primarily a function of exploration in breadth, in line with results obtained in Chapter 6. It can be hypothesized that scoring high on openness would probably apply to most searchers because being open to new experiences is probably at the heart of the willingness to explore various alternatives. Scoring high on neuroticism, however, would especially apply to those searchers that Côté and Schwartz (2002) would label characterological. These are the maladjusted searchers who are driven by unrealistically high standards (Côté & Levine, 2002), or who would be driven by maladaptive perfectionism (cf. Soenens, Elliot, et al., 2005).

Consolidators. The consolidators represent a rather new identity development cluster, which has not been empirically defined in previous research. Approximately 16% of our longitudinal sample belonged to this category. (The pathmakers, guardians, and searchers

comprised 32%, 30%, and 22%, respectively, of our sample.) Their main identity work appeared to be evaluating and consolidating the current identity structure. They scored the highest on commitment making, accompanied by the strongest upward trends across time. Exploration in breadth, however, was the lowest as compared to the three other clusters. However, exploration in depth was high at the onset of the study and strongly increased – much stronger than for the other individuals – during the first two years of emerging adulthood. Identification with commitment was the highest and remained rather stable over time. In sum, whereas the initial level and rate of change in exploration in breadth on the one hand correspond to some extent with the guardian category, the initial level and rate of change in identification with commitment on the other hand correspond to some extent with the pathmaker category. However, the high initial levels of commitment making and exploration in depth, coupled with the steep increases in both variables, are unique to the consolidator class.

When looking at the solution with three instead of four identity trajectory classes in the LGCA-analyses, no distinct consolidator trajectory class could be substantiated. When cross-tabulating the solution yielding three identity trajectory classes with the solution yielding the four identity trajectory classes, the consolidators overlapped to a substantial degree with the guardian trajectory class of the three-class solution, as demonstrated in Table 9.3. This could mean that the consolidators are a developmental subtype of the guardians. For instance, Archer and Waterman (1990) described a certain subcategory of foreclosures - open foreclosure - that to some extent resemble our consolidators. They are described as a group of adolescents who have committed themselves (i.e., high on commitment making) without demonstrating a genuine interest in other options (i.e., low on exploration in breadth) but they are characterized by a flexible orientation (cf. high on exploration in depth). However, they appear so secure in their choices (i.e., high on identification with commitment) that they demonstrate virtually no curiosity for other possible identity options.

Table 9.3

Cross-Tabulation of the Four Identity Class Solution With the Three Identity Class Solution

	Four Classes (*n*)			
Three Classes (*n*)	Searchers (86)	Guardians (119)	Pathmakers (127)	Consolidators (64)
Searchers (110)	**100.0**	17.6	2.4	0.0
Guardians (147)	0.0	**72.3**	0.8	**93.8**
Pathmakers (139)	0.0	10.1	**96.8**	6.2

Note. Cells contain percentages. Values indicate the percentages within columns.

The difference between the consolidators and the other identity trajectory classes becomes more pronounced when looking at the overlap with the adjustment trajectory classes. All individuals of the consolidator trajectory class overlapped with the increasing adjustment or the optimal adjustment trajectory class. Conversely, no one from the consolidators belonged to the stable maladjustment class.

With respect to the parenting characteristics, this trajectory class showed the highest scores on responsiveness and autonomy support and the lowest score on psychological control. Viewed in light of the reciprocal relationships between parenting and identity described earlier, it is plausible that, first, these individuals have very supportive and respectful parents that are empathic and understanding. Such a family background could be an important mechanism of action explaining why they handle identity issues the way they do and why they, for instance, address identity issues differently than the searchers do. Second, such a consolidating approach towards identity questions probably influences parents to a significant degree. Due to the fact that parents see that these adolescents are very certain concerning their choices made in areas such as education and friendship (and that their commitment to these choices even strengthens), instigates them to support their children intensively in the paths they have chosen. Similarly, the relative lack of exploration in breadth could explain why psychological control is the lowest in this category of individuals. In sum, parents probably perceive these adolescents as well underway on the route to adulthood because they have made very determined choices about which they have very little doubts left. It remains to be investigated, however, which strategy in emerging adulthood – searching or consolidating – proves to be most fruitful in the long term.

Regarding the personality correlates of the consolidators, they scored as low on neuroticism and as high on extraversion as both the pathmakers and the guardians. Their score on neuroticism, however, furthered decreased at Waves 3 and 5 and, consequently, they scored significantly lower on neuroticism than both the guardians and the pathmakers who continued to function at the same level of neuroticism throughout the study. In Chapter 6, we concluded that mean-level changes in personality were limited, with only Neuroticism evidencing some marginally significant decrease. The results discussed earlier emphasize the need to study personality development from a person-centered approach with a focus on meaningful subgroups (Caspi et al., 2005). Finally, the consolidators as a group scored the highest on agreeableness and conscientiousness and - as could be expected given their low score on exploration in breadth - as low on openness as the guardians.

Collectively, the additional analyses reported in this chapter indicated that the consolidator class indeed represents a distinctive trajectory of identity development in terms of

its associations with parenting and personality. The precise nature and meaning of this new identity trajectory, however, will have to be elucidated in future research.

10

Identity Formation in Emerging Adulthood:

Looking Forward

First, the present chapter focuses on practical implications. We hope that the results presented in this dissertation will not only speak to identity scholars but also to practitioners and counselors who deal with adolescents and emerging adults. Second, several important limitations of the present dissertation are described, along with various suggestions for future research. Three important research avenues are outlined in more detail.

Practical Implications

Throughout this dissertation, various practical guidelines were provided. The present section aims at bundling them and merging them into one coherent set of guidelines. First, a short overview of important previous intervention efforts will be presented.

Previous Intervention Efforts

Interventions represent efforts to promote healthy identity development in late adolescents and emerging adults in need of structure and guidance. Interventions have as their primary objective to facilitate movement from a less complete and coherent sense of identity to a more synthesized and workable identity in order to reduce confusion and uncertainty (Josselson, 1994). As highlighted in Chapter 1, such interventions might be most relevant in contemporary societies that lack structure and guidance on which to rely in forming a sense of identity (Côté, 1996, 2000; Schwartz, 2001). Such an integrated sense of identity is crucial in a person's life because it helps to unify the various aspects of one's life and everyday experiences, which provides a sense of direction and meaning (Schwartz, Kurtines, et al., 2005). This sense of identity also determines, at least in part, the outcomes on later developmental tasks. Consequently, some individuals who are challenged by this identity formation task, combined with the challenges present in the society as a whole, might benefit from certain interventions aimed at helping them to work through this process of identity formation.

A limited number of research efforts focused on interventions in identity formation. These efforts mainly targeted general skills such as decision making, social perspective taking, and problem solving skills (Enright, Olson, Ganiere, Lapsley, & Buss, 1984; Ferrer-Wreder et al., 2002; Markstrom-Adams, Ascione, Braegger, & Adams, 1993). Results suggested that identity processes, such as generating identity alternatives, can be manipulated and enhanced by way of intervention. However, the relative lack of maintenance of intervention gains on a long-term basis, as reported by Ferrer-Wreder et al. who focused on a sample of marginalized youth, might signal that intervention in and of itself is insufficient unless follow-up efforts are undertaken to strengthen long-term effects. Indeed, through increasing problem-solving skills

via participatory learning strategies, Ferrer-Wreder et al. only found short-term positive effects on identity indices (e.g., a decrease in the normative or foreclosed identity style).

In these previous intervention efforts, identity processes in general and identity exploration in particular were promoted using self-construction oriented problem-solving and decision-making processes (Schwartz, Kurtines, et al., 2005). In contrast, Schwartz (2002b) and Waterman (1993) viewed identity exploration also as an emotionally focused process, involving a discovery-based or affective examination of different identity options in terms of how well they resonate with one's inner self. Feelings of personal expressiveness might emerge as a result of the deep satisfaction that one experiences when engaged in activities that are reflective of one's unique potentials and needs. Self-construction and self-discovery represent independent but overlapping processes in this regard (Schwartz, 2002b). Self-construction refers largely to rational consideration of externally presented alternatives, whereas self-discovery refers largely to an intuitive sense of fit between identity alternatives and one's true self (Schwartz, Kurtines, et al., 2005). Schwartz, Kurtines, et al. have integrated the self-construction and the self-discovery approach through participatory learning strategies into the identity intervention field. They found that generation of alternatives was specifically affected by cognitively focused strategies, whereas personal expressiveness was specifically affected by emotionally focused strategies. In sum, using a combination of both strategies might be beneficial given that an increase in problem-solving skills and in the awareness of one's unique potentials is likely to emerge.

As is apparent from our results, guidelines for intervention will be divided primarily into two major categories, that is, guidelines targeting the developing individuals themselves and guidelines directed at the micro-context surrounding the developing individuals. Josselson (1994) also stated that two avenues of intervention are possible: the individual and the social context. Depending on the way individuals handle identity issues – or on the identity class they are situated in -, several barriers in identity formation may occur. They might be more "quantitative" in nature (such as a lack of exploration or too much exploration, or a lack of commitment making or a commitment made too fast) or more "qualitative" in nature (such as a commitment which was not made in accordance with inner values and needs). We will return to these problems in the next sections.

Interventions Aimed at the Individual

Importantly, none of the statuses or the identity trajectory classes are universally unhealthy (Josselson, 1994). Any of the identity status options could be adaptive given the right

personality configuration and social circumstances. Further, it is important that interventions should not be instituted for the purpose of undermining existing identity commitments, even those formed in a non-reflective or foreclosed manner. More important than the manner in which an identity commitment was formed, is the extent to which it is serving adaptive functions. Where a commitment is functional for an individual, there is a natural, and appropriate, resistance to efforts to undermine it. However, if at some future time a commitment is no longer functional, techniques can be used to find worthy alternatives that will provide a basis for commitment (Waterman, 1994).

Moratorium or searchers. For some searchers in our sample, the freedoms of the university life apparently proved to be too much for them to handle. Instead of exploring in an organized, systematic, or focused manner, they seemed to be drifting in their approach to identity issues. Consequently, their continual explorations were associated with experiences of disappointment, depressive symptoms, and insecurity. However, for other searchers of our sample, these explorations probably were illuminating for self-understanding and self-discovery. In other words, a continuous search for identity options may represent a source for instability or even disruption for emerging adults at the time it is undertaken (Orlofsky, Marcia, & Lesser, 1973), but may be beneficial for others. However, it seems important – as some of our results suggest - that exploration in breadth does result in firm commitment making at some time in emerging adulthood. Baumeister et al. (1985) made a distinction between two kinds of identity crisis, that is, identity deficit and identity conflict. An identity deficit arises when the person is struggling to make commitments due to an inadequately defined self or set of values. These adolescents might engage in protracted self-questioning to seek new sources of meaning and value. Important causes of this sort of identity crisis might be inconsistent, intrusive, and ambivalent parenting, in line with hypotheses upheld throughout the present dissertation.

There are some people for who searching becomes a way of life (i.e., characterological moratorium) rather than a path toward identity (i.e., transient moratorium). For these former people, it becomes dangerous when guilt and anxiety become so overwhelming that the clear thinking necessary to make choices becomes clouded or blocked. Consequently, the presence of excessive depressive symptoms or anxiety in searchers might signal the presence of pathological (identity) processes that need a remedy of some sort (Josselson, 1994). It has to be noted, however, that most emerging adult moratoriums are doing what they – from a developmental point of view – ought to be doing, that is, launching themselves into the exploration process (Marcia, 1994). Hence, what is needed is a counselor who is patient and responsive and who makes clear to the adolescent that the process he or she is undergoing,

although painful, is likely to be productive in the short or long term. So, a thorough exploration of alternatives needs to be valued, rather than disparaged, although it is likely that some individuals will need more guidance on the road (cf. the notion of structure accommodation as labeled by Marcia, 1994).

Fuqua and Hartman (1983) stipulated that identity problems might be related to general indecisiveness or indecision in particular (Germeijs & De Boeck, 2002, 2003). Osipow (1999, pp. 147-148) defines both concepts as follows: "Indecision is a developmental phase through which individuals may pass on their way to reaching a decision. It is a state which comes and goes over time as a decision is made […]. Indecisiveness is a different process. Whereas indecision is a state that is normal in human development, indecisiveness is not an ordinary part of human growth and development, but is, instead, a personal trait which generalizes across situations demanding decisions". Referring back to the distinction between transient and characterological moratoriums, it seems plausible that the latter category might be characterized by a rather stable trait of indecisiveness which generalizes across situations and which leaves individuals rather unable to make commitments. These individuals experience fear when engaged in decisional processes and, consequently, they avoid making decisions and turn to rumination or procrastination (Milgram & Tenne, 2000; Rassin & Muris, 2005). Such a chronic decisional procrastination is likely to produce feelings of loss of control of one's life and, consequently, maladaptive functioning (Milgram & Tenne, 2000). The distinction between indecision and general indecisiveness has clear implications for counseling (Heppner & Hendricks, 1995; Petit, 2005). Counseling with an individual who is characterized by indecision might best focus on the decisional process itself (e.g., by means of providing greater insight in the different alternatives and in one's own set of values and needs). In contrast, general indecisiveness requires a more intensive and personal counseling format with specific attention to one's personality, anxieties, and direct environment (Germeijs & De Boeck, 2002; Germeijs, Verschueren, & Soenens, in press; Peng, 2001).

Foreclosure or guardians. The results of our dissertation showed that most guardians functioned as well as pathmakers did during the first two years of emerging adulthood in terms of well-being and self-esteem. Apparently, the commitments made by guardians and pathmakers, regardless of the amount of exploration in breadth, proved to be functional for most of them. Should we nonetheless intervene in the guardian trajectory class because these commitments were made probably with limited prior exploration in breadth? Following Waterman (1994), we think not. In contrast to some moratoriums who struggle at the cost of anxiety, foreclosures, however, do tend to trade freedom and choice for certitude and escape

from doubt and ambivalence. These characteristics might become problematic when they clash with the self-awareness and self-realization needed on the path towards an adult identity (Josselson, 1994). Only when later in life it indeed becomes clear that the quality of commitments do differ between guardians and pathmakers, intervention efforts might be warranted.

For some foreclosures, the very fact of exploration itself might be threatening and not so much what is explored (Marcia, 1994). Marcia, Waterman, Matteson, Archer, and Orlofsky (1993) hypothesized that a primary reason for this experience of threat might be the guilt and anxiety associated with questioning parental values. Consequently, the rule with foreclosures who report problems is to proceed gradually because a danger might be to be too successful too quickly (Marcia, 1994). If foreclosures start to explore, they start questioning the internalized figures or standards. Consequently, if they depart from these introjects, they risk having no internalized standard to judge one's worth. This feeling of emptiness might be more counterproductive than the initial foreclosed state (Marcia, 1994; Schwartz, 2001). A cautious disequilibration of the internal structure coupled with the installment or the discovery of a new one is warranted in these individuals.

Foreclosures might enter identity-related counseling when problems are anticipated with regard to the implementation of an established commitment in adult life (Waterman, 1994). A feeling might arise that the chosen goal or value might not prove to be as fulfilling as had been believed originally (Raskin & Waterman, 1994). Foreclosures strongly uphold and internalize social or external values and beliefs with the possible danger of frustrating their own needs and values. According to self-determination theory (SDT; Ryan & Deci, 2002), the degree to which this process of internalization functions successfully can vary considerably. At the low end of the continuum we find external regulation and at the high end we find integrated regulation. In between these two poles, we find introjected regulation and identified regulation, respectively.

In the case of introjected regulation, the enactment of the behavior does no longer require external contingencies, as is the case for external regulation. However, the behavior is now dependent on internalized contingencies that have not been accepted by the person as his or her own. People act out of a sense of internal pressure and obligation; they still feel as if they have no other choice than to do the activity. In contrast, in the case of identified regulation, people feel volitional in carrying out the behavior because they experience the behavior as a reflection of their own values (Vansteenkiste, 2005). Foreclosures or guardians entering counseling because the chosen goal or value might not prove to be as fulfilling as had been originally assumed, might be characterized by introjected regulation. The taking in of parental or societal

values by these individuals probably did not imply that this psychological material, although part of one's identity, has been brought in alignment and harmony with one's inner values and needs.

In intervention efforts, it might be crucial that some foreclosures are learned to move beyond acting out of externally or internally pressuring forces and that they are learned to act more upon their inner integrated values and needs. One important route to this self-discovery might be a thorough exploration of one's inner self, values, and needs. This might imply that self-construction processes must have been set in motion to some extent before self-discovery interventions are likely to be effective (cf. Schwartz, 2002b, who stated that self-construction processes might serve the path to self-discovery).

Diffusion or drifters. Some of these individuals might be considered appropriate candidates for counseling or therapy and could benefit from rather directive intervention efforts (Schwartz, 2001). In general, however, gentleness and patience are bound to be key aspects of any intervention strategy because, for many disturbed diffusions, the intervention process will be a long one with a focus on structure building. The more disturbed diffusions might come in different sorts in emerging adulthood. Marcia (1994), for instance, made a distinction between the schizoid loner and the empty vessel. Archer and Waterman (1990) further elaborated on these subtypes within the diffusion status and distinguished among various subtypes, such as apathetic identity diffusion and alienated identity diffusion. Both subcategories not only lack strong identity-related commitments, they also show no interest in forming them. Such nonchalance, however, might mask an underlying uncertainty about successfully addressing identity-related issues.

As demonstrated in Chapter 4, it is indeed necessary to distinguish among various types of diffusions before undertaking psychotherapy or counseling. A distinction was made between diffused diffusion – or the more disturbed diffusions – and carefree diffusions. Carefree diffusions seldom appear for therapy or counseling on their own initiative because their current state of avoiding commitments and exploratory identity work on a regular basis appears quite functional for them in the college context. However, as stated in Chapter 4, it remains to be investigated if carefree diffusions continue to function well as they leave university and enter the adult world with its demanding tasks. Although they might be socially skilled, they possibly failed to live up to the best of their potentials in emerging adulthood, which could render them rather unprepared for some adult responsibilities and tasks (Marcia, 1994).

Interventions Aimed at the Micro-Context

Social intervention necessitates an analysis of the role of the context in supporting or hindering identity formation (Josselson, 1994). Marcia (1994), for instance, paid a lot of attention to the therapist – client relationship. He strongly emphasized the encouraging and supportive aspect of psychotherapy (Marcia, 1989b). The therapist has to establish a protective and nurturing context – Raskin and Waterman (1994) referred to listening without making judgments - in order to facilitate exploration and commitment in the client.

Family context. Our results demonstrated that parents, through the use of enabling or supportive and constraining or intrusive interactions, can shape and guide emerging adult identity formation (Hauser, Powers, Noam, & Jacobson, 1984; Papini, 1994; Tokar, Withrow, Hall, & Moradi, 2003). Consequently, intervention efforts should be directed at family contexts because adolescent explorations and commitments are facilitated by family contexts that are emotionally connected, autonomy supportive, and flexibly adaptive. Emotionally detached, psychologically intrusive, or enmeshed families and rigid or chaotic families are unable to provide emotional security and to adapt to their maturing children's developmental needs of exploration and commitment to life options. The tendency for adolescents in enmeshed families is to retreat to the safety and security of parent-child relationships, whereas the tendency for adolescents from detached and rejecting families is to continue their aimless psychological wanderings without making psychological commitments (Papini, 1994). The latter mechanism was partially demonstrated in our analyses in Chapter 7 on the link between psychological control, commitment, and exploration.

Apparently, the key to optimizing adolescent or emerging adult identity development is not to focus solely on the developing individual him- or herself but on directing strength-promoting interventions towards the family as a whole. Consequently, clinical family therapy or family educational programs might be warranted in some cases of problematic identity formation. Establishing optimal levels of psychological connectedness to each other with mutual respect and understanding should be one of the primary targets of these programs (Tokar et al., 2003).

School and college context. Dreyer (1994) focused on the educational environments and curricula that could contribute to the achievement of a functional identity. Dreyer stated that the teachers' responses to the various roles adolescents experiment with have much power to aid or interfere with the process of identity formation. For the college environment to contribute to healthy identity development, it is crucial that students feel that their schoolwork represents an investment of their own feelings and talents so that they can assume ownership over their own

work (Sierens, Soenens, Vansteenkiste, Dochy, & Goossens, 2005). Knowledge, critical thinking, and responsible choice are important elements that need to be emphasized by school or college environments and teachers. Teachers must treat students with respect and as competent individuals acknowledging that the exploratory behavior of youngsters may not always be error-free or successful.

Sierens et al. (2005) argued in favor of the arguments put forward by Dreyer (1994) and provided a conceptual analysis of teaching styles starting from the parenting literature and SDT. Although they focused specifically on the relationship between teaching styles and self-regulated learning, their ideas are very relevant to the process of identity formation and how teachers might impact on this process. In short, they state that, in line with Grolnick, Deci, and Ryan (1997), teachers need to score high on three dimensions in order to improve optimal functioning in their students: involvement, provision of structure, and autonomy support. These three dimensions are important in all interpersonal relationships (e.g., parent – child and employer – employee; Vansteenkiste, 2005). Involvement is defined as the degree to which teachers are caring and warm persons that show an active interest in their students. Provision of structure refers to the degree to which clear expectations are communicated to the students and constructive feedback is provided. Finally, autonomy support –in line with our view on autonomy support as an aspect of parenting– is viewed as the provision of age-appropriate choices and the provision of sufficient freedom to students to strive for personally relevant goals in contrast to using over-controlling or intrusive techniques. The interplay of these three dimensions probably gives rise to acting in a self-determined or self-regulated fashion on the part of students.

In line with this theorizing, Soenens and Vansteenkiste (2005) found that the degree of teachers' autonomy support was positively associated with identification with commitment and exploration in depth in the domain of vocational identity. Further, strong linkages were reported between adolescents' self-determined motives for job-searching and both exploration in depth and identification with commitment. These findings imply that, in line with Flum and Blustein (2000), the degree to which adolescents engage in their search for a job with a sense of volition and self-determination is fundamental to their vocational identity. In sum, teachers and educative contexts in general should be stimulated to create an optimal environment that improves the experiences of self-determination on the part of the students. Enhancing the subjective experience of self-determination, in turn, might have an important impact on identity formation.

Limitations

The majority of the emerging adults in our sample were female, which seriously limits the generalizability of our results. However, as mentioned for instance in Chapter 8, many of the primary results obtained were not influenced by gender or were the same when all males were deleted from the sample and the analyses were conducted again on an exclusively female sample. This means that, as expected, the structural relations and developmental associations and trends identified in our dissertation are probably not that different for men and women. This invariance across gender is in line with Kroger (2003) who concluded that, with the exception of domain salience (Fadjukoff et al., 2005), there has been little evidence of gender differences regarding questions of identity structure, domain salience, or developmental process. Of course, this hypothesis has to be tested on a balanced sample in terms of gender, drawn in a longitudinal study.

A similar reservation can be made regarding the use of college students only as participants. Few studies to date directly examined the relationship between late adolescent or emerging adult identity formation and different vocational settings (Kroger, 2000a). Consequently, little is known about identity formation outside the college setting. Research findings to date are rather mixed (Archer & Waterman, 1988; Morash, 1980; Munro & Adams, 1977). Whereas some studies found more individuals being identity achieved among late adolescents who were at work compared with those attending college, other studies found college students to be more advanced in identity development than those at work (Schwartz, 2005). Different mechanisms can be hypothesized to be at work that could explain some of these mixed results (Yoder, 2000). Working adolescents may be more pressured by the concrete working context to make life decisions (or, in some cases, to foreclose) on identity-related topics compared to adolescents or emerging adults who are still in school. Conversely, working adolescents may be granted less time to indulge themselves in extensive explorations of various identity alternatives because they cannot benefit from the psychosocial moratorium provided by the college setting. Clearly, future research is required to empirically test those competing hypotheses (Schwartz, 2005).

A third reservation can be raised concerning the sole reliance on self-report questionnaires that reflect only the viewpoint of the emerging adults themselves. Several variables included in this research project (e.g., parenting and adjustment), however, invite a multi-informant approach. Hence, the initiation of a new longitudinal research project using multiple informants is needed to check whether the results obtained in our dissertation are robust across these different methods of data collection. Recent research conducted in our research group (e.g.,

Soenens, Elliot, et al., 2005) has demonstrated that using both parent and adolescent reports as indicators of a parenting construct yields results that are highly similar to the use of adolescent self-reports only. Further, adolescent self-report may be the most valid measure to assess dimensions of parental treatment, due to the subjective nature of this experience (Barber, 1996). Nonetheless, future research should rely on multiple informants. For instance, it could be the case that the observed relationship between identity formation and adjustment could be inflated due to a tendency to answer in a socially desirable fashion (e.g., the foreclosures or guardians who probably value social norms very highly).

Suggestions for Future Research

Three other important avenues for future research (i.e., optimizing the assessment of dimensions of identity formation, identity formation in younger and older age cohorts, and identity formation in emerging adults who face additional, non-normative challenges) are described in more detail. Attempts to investigate these research questions are underway or will be undertaken at our research center, taking into consideration the need for more balanced samples in terms of gender and educational background and the use of multiple informants.

Optimizing the Assessment of Dimensions of Identity Formation

Important limitations of the identity questionnaires used. The fact that we used existing questionnaires to operationalize the identity formation process at the dimensional level had several disadvantages; by doing so, we inherited the problems accompanying these questionnaires. Although we tried to remedy some of these shortcomings the best we could, they have to be discussed in more detail in order to remedy them in future research.

First, we used the Ego Identity Process Questionnaire (EIPQ; Balistreri et al., 1996) and the Utrecht-Groningen Identity Development Scale (U-GIDS; Meeus, 1996) together in all empirical chapters. The identity dimensions assessed by these questionnaires were thought to constitute the core and substance of identity formation in emerging adulthood. The content domains assessed by both questionnaires, however, differ to some degree. The two questionnaires have in common that they measure both the ideological domain and the interpersonal domain of identity formation (Goossens, 2001). Whereas the ideological domain is represented in the EIPQ by the specific content domains of education and future profession, politics, religion, and values, the U-GIDS assesses only the specific content domain of education and future profession. Similarly, whereas the interpersonal domain is represented in

the EIPQ by the specific content domains of friendship, intimate relationships, family, and gender roles, the U-GIDS assesses only the specific content domain of friendship.

These content differences partially led us to mainly focus on dimensions of identity formation at a global or overall level. In other words, identity scores within the different content domains were used as indicator variables in order to capture the structural dimensions of identity formation that would be valid across content domains. Although frequently used (Goossens, 2001; Meeus et al., 1999), this approach represents a somewhat limited view on identity formation because identity may take on a different form across domains (Coleman, 1978; Grotevant, 1987; Kroger, 1986, 1988; Schwartz, 2001).

Second, some of the identity dimensions that were assessed by the two questionnaires mentioned earlier were based on somewhat narrow definitions of the intended constructs, particularly in the case of identification with commitment. A detailed inspection of the U-GIDS items indicates that they focus on the degree to which chosen commitments provide self-confidence and certainty concerning the future. Although these are indeed crucial aspects of identifying oneself with the commitments chosen, we hypothesize, in line with SDT (Deci & Ryan, 1985) and eudaimonistic identity theory (Waterman, 1984, 1993), that other important factors also drive the degree to which chosen commitments are being experienced as worthy and satisfying. The degree to which the commitments are reflective of the inner self and correspond to the inner wishes, desires, and needs (i.e., the degree to which chosen commitments and activities are experienced as being personally expressive) probably constitute important processes in the pathway to adult identity development. Hence, identification with commitment needs to be expanded with more motivational-affective items to tap the full range of this construct.

Third, the identity model laid out in this dissertation only captures a very limited part of the rich process that is identity formation. Partially based upon the work of Erikson (1950, 1968), three distinct levels of identity, that is, ego identity (i.e., the innermost, silent, and unconscious process underlying identity formation), personal identity (i.e., the goals, values, and beliefs one has in interaction with the world), and social identity (i.e., group identifications and one's assigned and chosen place in the world of social groups; Phinney, 1990; Tajfel, 1982), are distinguished in the literature (Schwartz, 2005). The bulk of identity status research has attended to personal identity. This is also the case for the research presented in this dissertation. However, only a limited aspect of the processes at work in one's personal identity was captured in the identity formation model presented in this dissertation.

Temporal-spatial continuity may represent a crucial component of the identity development process that is not explicitly captured in our dissertation. Because identity is assumed to operate across multiple content domains (Goossens, 2001; Grotevant, 1987; Schwartz, 2001), consistency across domains and over time is an important aspect of identity development (Blasi, 1988; Blasi & Glodis, 1995; Dunkel, 2005). The degree to which one experiences oneself as an integrated and unified entity across different commitments is an important component of functioning (Cross, Gore, & Morris, 2003; Suh, 2002). Indeed, such continuity has been identified as an integral component of identity formation in various contributions (e.g., Blasi, 1988; Côté & Levine, 1987, 1988; Erikson, 1968; van Hoof & Raaijmakers, 2002). Sheldon and Kasser (1995) further emphasized that it is crucial to consider not only the coherence of different elements into a personality system, but also the congruence of those elements with inner needs and values. Luyckx, Schwartz, Soenens, and Goossens (2006) indeed found identity integration (as an index of temporal-spatial continuity; Campbell et al., 1996; Campbell, Assanand, & Di Paula, 2003; O'Brien & Epstein, 1988) to be a crucial mediating mechanism in the pathway from identity commitments to adjustment. Further research on this topic is warranted.

Solutions to these problems: The development of a new questionnaire. In an attempt to remedy some of these shortcomings, we are developing a new questionnaire that takes into account some of these problems. This new questionnaire aimed at assessing the four identity dimensions that were hypothesized to play a crucial role in the identity formation process of late adolescents and emerging adults. For the initial testing of the new identity questionnaire, content domains were selected that were hypothesized to have particular relevance for college students or emerging adults and that would allow meaningful variability of choice for the four identity dimensions assessed (such as education and future profession, friendship, and leisure time activities). When the identity dimensions need to be assessed in younger age cohorts, specific cultural-ethnic groups, or special populations, other content domains can be included and some content domains can be deleted if necessary (Fadjukoff et al., 2005). One important advantage of assessing the same content domains with regard to all four identity dimensions is the opportunity it provides to check for congruence or incongruence across content domains with regard to development and change in these identity dimensions and to focus on the actual ways in which individuals integrate identity domains (Grotevant, 1993). With regard to the identity statuses, substantial intra-individual incongruence was observed in status classification across the different content domains (Fadjukoff et al., 2005; Goossens, 2001), so further research at the dimensional level is warranted.

An initial version of this questionnaire was pilot tested in a sample of psychology and educational sciences freshmen in October 2005 (*N* = 355). In creating the item pool of this preliminary version, item stems from the EIPQ and the U-GIDS were used. Further, some item stems from the revised Identity Style Inventory (ISI3; Berzonsky, 1992; Berzonsky & Kuk, 2000) and the revised Extended Objective Measure of Ego Identity Status (EOM-EIS-II; Bennion & Adams, 1986) were revised to assess exploration and commitment, as opposed to a particular identity status or style. Finally, to assess the new affective-motivational items of the subscale Identification with commitment, item stems from the Personally Expressive Activities Questionnaire (PEAQ; Waterman, 1993) were used. Importantly, all items for the four identity dimensions were assessed for the same content domains. Initial results on reliability, factorial validity, and external validity were very encouraging (and in line with the results described in this dissertation) and were partially replicated in a second study in November 2005 focusing on sophomore students in Psychology and Educational sciences (*N* = 264). More studies in different samples using this new questionnaire are currently underway in order to assess the validity and reliability (e.g., test-retest reliability) of this new questionnaire.

Identity Formation in Younger and Older Age Groups

As indicated in Chapter 8, a long-term follow-up of identity issues in emerging adults is needed well into adulthood (Kroger, 2000a; Schulenberg, Sameroff, et al., 2004). Several studies have already addressed identity formation in adults (Cramer, 2004; Fadjukoff et al., 2005; Marcia, 2002; Pulkkinen & Kokko, 2000) and most of them converge on the finding that development and change continue to be predominant aspects of identity formation throughout adulthood. These research reports focused predominantly on the identity statuses and, consequently, only on the dimensions of exploration in breadth and commitment making. Chapter 5, however, indicated that both dimensions of the commitment evaluation cycle (i.e., exploration in depth and identification with commitment) constitute important developmental core aspects of identity formation in the transition to adulthood that probably should not be neglected in research in adults. Successive identity decisions and evaluations continue to be inevitable into adulthood (Marcia, 2002).

How one progresses through the core developmental tasks of late adolescence and emerging adulthood echoes into adulthood and partially determines how one will cope with the opportunities and pitfalls one encounters in adulthood (Arnett, 2000; Marcia, 2002; Obeidallah, Hauser, & Jacobson, 1999). This continuity of developmental tasks - with earlier developmental tasks influencing the attainment of success or failure on later developmental tasks - already

formed the core of Erikon's (1968) epigenetic life-span model. Marcia (2002) describes several possibilities how identity formation in late adolescence could relate to later developmental tasks, such as intimacy versus isolation in early adulthood or integrity versus despair in late adulthood (Kroger, 2000a). Marcia stated that if individuals have been non-exploratory in their approaches to identity in young adulthood, they probably are not going to risk their peace of mind in older age. Clearly, how one evolves through the identity formation process of late adolescence (i.e., exploration-oriented versus foreclosed) could lay the foundation for how one evolves through later life.

In this respect, Obeidallah et al. (1999) envisioned three competing long-term developmental trajectories that reflect three pathways of general psychosocial experience. First, the *continuity effect*-hypothesis states that individuals who experience optimal functioning in adolescence probably will move on to experience optimal outcomes in adulthood, due to the way in which they handled certain developmental tasks in adolescence. Second, the *rebound effect*-hypothesis states that initial outcomes associated with developmental match could be short-lived; experiencing a good fit during adolescence would not necessarily lead to optimal outcomes in adulthood. Third, the *sleeper effect*-hypothesis states that the developmental work that has been undertaken in adolescence is not accompanied by psychosocial benefits in adolescence, that is, in the short term. Instead, optimal psychosocial functioning will become evident in later life, when one has to draw on individual resources to cope with the challenges of adulthood. Those individuals with a developmental mismatch may not experience lowered psychosocial outcomes in their adolescent years. Instead, developmental mismatch in adolescence may create vulnerabilities that undermine later optimal development.

These three hypotheses shed a different light on some of our findings and emphasize the need to view identity formation within a long-term developmental framework. Exploration in breadth was accompanied by simultaneous depressive symptoms, both cross-sectionally and longitudinally. Consequently, several chapters ended by concluding that a continual exploration in breadth in late adolescence and emerging adulthood might signal rumination and, hence, could signify a rather problematic developmental position for some individuals. The continuity effect-hypothesis would imply that this simultaneous detrimental effect of exploration in breadth in emerging adulthood continues to exist during adulthood. However, the sleeper effect-hypothesis would imply that the beneficial effects of exploring in breadth in emerging adulthood would mainly come to the fore when individuals have reached adulthood, because these individuals would be able to use more psychological resources than those who did not explore in breadth during late adolescence and emerging adulthood. This line of reasoning

dovetails nicely with Arnett's (2000) thinking on this topic. He stated that explorations in emerging adulthood are probably beneficial in the long run because they allow individuals to obtain a broad range of life experiences before taking on enduring adult responsibilities. Entering adulthood represents a closing of doors and the end of a sense of wide-open possibilities (Arnett, 2004). Further, he emphasizes that this does not mean that these explorations are experienced as enjoyable at the time they are undertaken, as is clearly demonstrated in our dissertation.

Similar questions can be raised concerning some of our developmental trajectory classes. The continuity effect-hypothesis would imply that the consolidators would continue to demonstrate the best adjustment profiles in adulthood. The rebound effect-hypothesis, on the other hand, would imply that these beneficial effects would be short-lived. Long-term beneficial effects would be rather absent due to their exclusion of a broad identity exploration and experimentation in their college years, the latter being opportunities that are not likely to be possible during the thirties and beyond (Arnett, 2000). In short, a long-term follow-up is needed to assess how valid the conclusions drawn in the present dissertation are for individuals beyond the age boundaries examined.

Identity formation in mid-adolescence and in the transition to late adolescence also needs to be studied longitudinally (Kroger, 2000a). Up till now, there is a predominance of identity research that focuses on late adolescence or on emerging adults – the tail end of Erikson's identity stage (Schwartz, 2005) - so we partially remain in the dark about specific identity processes and identity development in younger adolescents. More specifically, little is known about the pathways that lead to the identity trajectories and changes described in this dissertation. Why do some emerging adults continue to explore various alternatives and do not arrive at making firm commitments (i.e., the searchers), whereas other emerging adults cease to explore in breadth various alternatives and already have a determined set of commitments (i.e., the consolidators)? In other words, what happened in the years before entering college (i.e., the period in life in which identity issues are first envisioned and undertaken) that could partially determine these developmental stances in emerging adulthood? Ideally, of course, intensive and long-term longitudinal studies need to be undertaken that follow different cohorts of individuals from early adolescence on into emerging adulthood and adulthood in general (Schwartz, 2005), enabling researchers to study identity development from a life-span perspective.

Emerging adults who are chronically ill have to deal simultaneously with normative developmental tasks characteristic for this age period and a substantial non-normative task, that is, coping with their chronic illness (Obeidallah et al., 1999). These two tasks will probably interact and influence one another (Seiffge-Krenke, 1998). Put differently, chronically ill adolescents and emerging adults are often confronted with a wide range of difficulties or stressors as they struggle with "normal" development while carrying the burden of living with an intrusive illness. This disability or chronic illness may cause a feeling of a flawed identity and thus interfere with the characteristics that are specific for emerging adulthood as a developmental stage (Fitzpatrick, 2002; Holmbeck, 2002).

This topic has been somewhat neglected in developmental research to date (Adams, Pill, & Jones, 1997; Hosek, Harper, & Robinson, 2002; Ryff & Singer, 2000), despite the fact that 10 – 20% of Western adolescents have to deal with chronic illnesses (Williams, Holmbeck, & Greenley, 2002). About 10% of chronically ill adolescents continually suffer from a severe illness that substantially impacts on developmental issues (Persoons, Luyckx, Desloovere, Vandenberghe, & Fischler, 2003; Scharloo et al., 1998; Wallander & Varni, 1998). For instance, Charmaz (1995) stated that chronic illness experiences could slow the process of forming a stable identity. Put differently, an identity avoidance orientation may come to the fore in some emerging adults who have to deal with a chronic illness (Hosek et al., 2002). Apparently, integrating the illness within the total developmental process of emerging adulthood represents a daunting challenge for some (Sayer, Hauser, Jacobson, Willett, & Cole, 1995).

On the other hand, the study of identity issues in chronically ill emerging adults could also shed light on the mechanisms that promote resilience and resistance from mental ill-being and developmental setbacks (Antonovsky, 1987; Ryff & Singer, 2000; Williams et al., 2002). For instance, Hosek et al. (2002) reported that several adolescents who were confronted with HIV infection experienced a sense of personal growth and heightened introspection as a result of confronting and exploring their illness. Some adolescents may feel strengthened through coping with the chronic illness and strictly adhering to a medical regimen and, hence, feel stimulated in their personal development (Gortmaker, Perrin, Weitzman, Homer, & Sobol, 1993). Similarly, Seiffge-Krenke (1998) found that diabetic adolescents perceived themselves as competent as their healthy peers in terms of the fulfillment of developmental tasks at hand and showed an overall positive transition to adulthood, resulting in resilient development (Hauser, 1999).

In line with this observation, most researchers agree that severe problems only occur in a subset of chronically ill adolescents (Howe, Feinstein, Reiss, Molock, & Berger, 1993). More specifically, it has been established that chronic diseases with a neurological background (e.g., epilepsy and spina bifida) show a greater risk than those with a non-neurological background (e.g., asthma and diabetes). Others put forward that the incidence and extent of psychosocial problems accompanying chronic diseases may depend on other characteristics of the chronic disease, such as severity or visibility (Seiffge-Krenke, 1998). Consequently, person-centered longitudinal research (Nagin, 2005) is greatly needed to focus on the variability within the group of chronically ill adolescents in order to investigate why adolescents with chronic diseases develop differently with regard to identity issues and psychosocial functioning (Hauser, 1999; Ryff & Singer, 2000).

At the end of this dissertation, we express the hope that future research, which uses improved questionnaires, explores age periods adjacent to emerging adulthood, and addresses the specific situation of young people who face multiple challenges, will continue to expand our knowledge of identity formation at the threshold of adulthood.

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